Department: Mechanical Engineering, KAIST

ME.40000-B Capstone Design I (창의적 시스템 구현)

<Automated System Design>

Instructor: Prof. Ki-Uk Kyung

Email: kyungku@kaist.ac.kr

Lecture: Fri 11:00~12:00
Lecture classroom: ME Building (N7) Room 2204
Practice classroom: Narae Practice Building (N25) Room 3330, 3332 (connected with Industrial Design B/D 3F)
(All lectures and presentations will be held in the lecture and all group meetings will be held in practice classroom or self-arranged space if there is no further notice.)

Credits: 3

Course Type: Project-Based Capstone Design (Team-Based)

Mentoring Professors:
Minkyun Noh(T1), Yong-Hwa Park(T2), Hae-Won Park(T3), Hongki Yoo(T4), Seibum Choi(T5)

TA:

Main TA:	Minjae Cho (micky97@kaist.ac.kr), Sein Lim (onsein@kaist.ac.kr), Jongseok Nam(tgb02388@kaist.ac.kr)
SW manager TA:	Hyeongjun Kim (kaist0914@kaist.ac.kr), Jeong Hyun Lee (joshualee@kaist.ac.kr)
HW manager TA:	Taegyu Song (thdxorb5515@kaist.ac.kr), Kyungmin Nam (twinhk@kaist.ac.kr),
Team Mentor TA:	Young-Rang Seo (syr0024@kaist.ac.kr), Moonkyu Jung (moonk127@kaist.ac.kr), Yonghoon Lee (ryyh33@kaist.ac.kr), Minho Lee (myno1126@kaist.ac.kr), Hyunsik Oh (ohsik1008@kaist.ac.kr

Course Description
This course is a project-based capstone design course in which students work in teams to design, build, and demonstrate a stand-alone automated system that accomplishes a given mission. An automated system in this course consists of sensors, actuators, mechanical structure (body), controller, and power source, integrated into a fully functional system capable of autonomous operation.
The mission will be introduced at the beginning of the semester and may be refined prior to the start of the course. As an example, the 2025 mission required each system to hit a bell located at the top of a vertical post.
Each team will propose, design, fabricate, integrate, and test their own solution. The course culminates in a competitive final demonstration among teams.

Course Objectives
By the end of this course, students will be able to:
- Design a complete automated system from concept to realization
- Integrate mechanical, electrical, and control subsystems into a unified platform
- Apply engineering design methodologies under real-world constraints
- Work effectively in multidisciplinary teams
- Communicate design intent, analysis, and results through reports and presentations
- Demonstrate a functional autonomous system that successfully completes a mission

Course Format
- Team-based project work (5–6 students per team)
- Weekly design reviews and technical discussions
- Hands-on prototyping, fabrication, and testing

- Instructor and TA feedback throughout the design process
- Inter-team competition at the end of the semester

Mission Overview

- The mission defines the functional goal of the automated system
- Detailed rules, constraints, and evaluation metrics will be announced at the beginning of the semester
  Example (2025): Hitting a bell at the top of a post
- The mission may involve constraints on size, weight, power, time, autonomy level, or safety

Team Formation

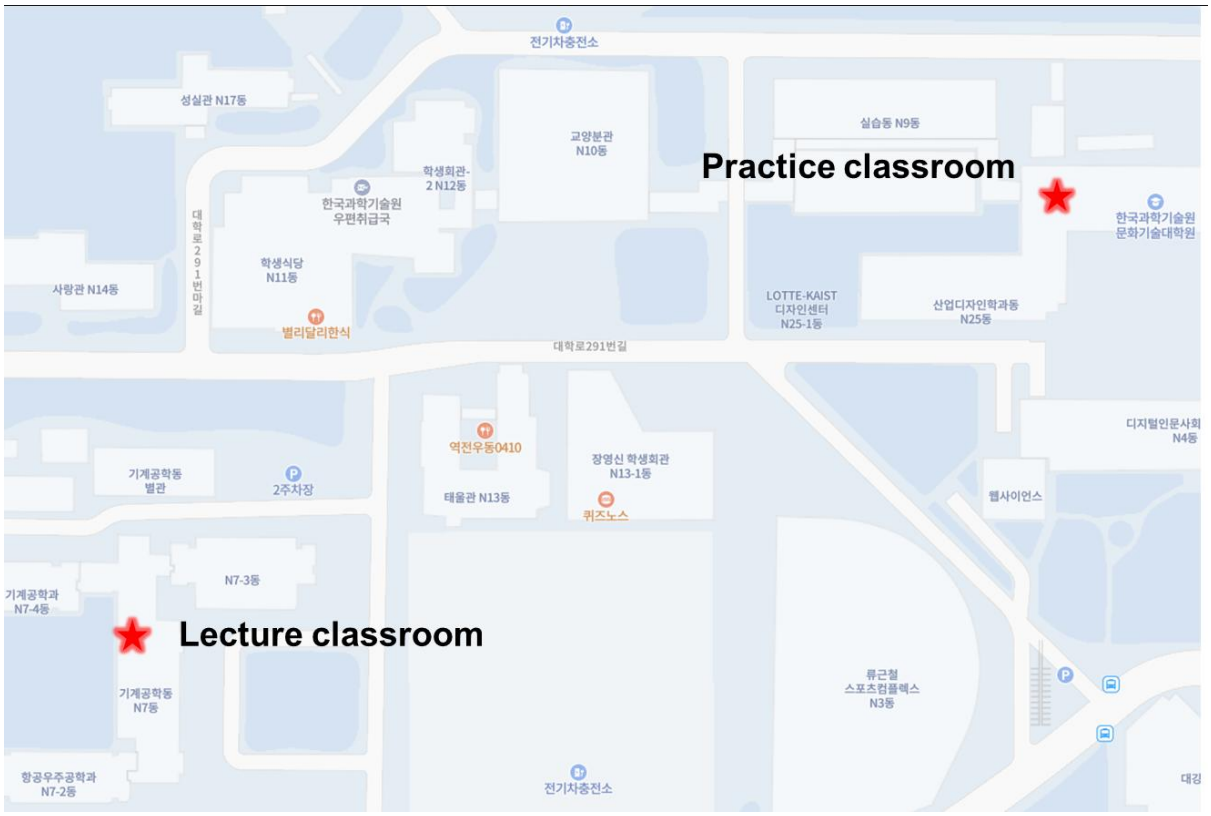
- Teams consist of 5–6 students
- Teams are expected to distribute responsibilities across:
- Mechanical design & fabrication
- Sensors & actuators
- Control & software
- System integration & testing
- Documentation & presentation

Assessment & Grading

- Attendance & Participation 10%
- Mid-term Review 15%
- Final Demonstration & Presentation 55%
- Final Report 10%
- Peer Evaluation 10%

Lecture Schedule

Week #	Date	Contents
1	**3/6**	**Course Introduction & Assigning Student Groups along with mentoring TAs and Professors**
2	**3/13**	**Software Introduction (09:00~12:00) @ lecture room**
3	**3/20**	**Hardware Introduction (10:00~12:00) @ lecture room**
4	**3/27**	**Group Meeting (11:00~12:00) @ practice room**
5	**4/3**	**Group Meeting**
6	**4/10**	**Group Meeting**
7	**4/17**	**Midterm Presentation**
8	**4/24**	**Midterm Exam Week (No Class)**
9	**5/1**	**Group Meeting**
10	**5/8**	**Group Meeting**
11	**5/15**	**Group Meeting**
12	**5/22**	**Group Meeting**
13	**5/29**	**Final Demonstration & Presentation**
14	**6/5**	**Final Report Deadline**
15	**TBD**	**Award Ceremony**
16	**6/19**	**Final Exam Week (No Class)**


Practice classroom
Lecture classroom
전기차충전소
성실관 N17동
교양분관
N10동
실습동 N9동
학생회관-
2 N12동
한국과학기술원
우편취급국
한국과학기술원
문화기술대학원
대학로291번길
학생식당
N11동
사랑관 N14동
벨리달리한식
LOTTE-KAIST
디자인센터
N25-1동
산업디자인학과동
N25동
대학로291번길
디지털인문사회
N4동
역전우동0410
장영신 학생회관
N13-1동
기계공학동
별관
2주차장
태울관 N13동
퀴즈노스
웹사이언스
N7-3동
기계공학과
N7-4동
기계공학동
N7동
류근철
스포츠컴플렉스
N3동
항공우주공학과
N7-2동
전기차충전소

3F: ME Narae Practice Building
기계공학과 나래실습동
ME Narae Practice Building
소
화
전

Practice classroom